Sri Lanka Institute of Information Technology

Faculty of Computing

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IT2130 - Operating Systems and System Administration

Year 02 and Semester 02

Lecture 10

Hard Disk Scheduling in Operating Systems

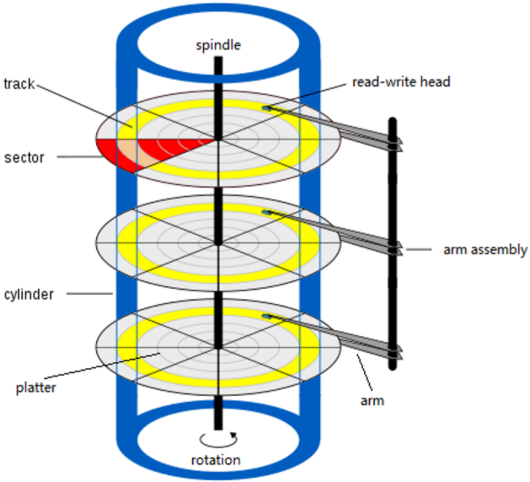
Lecture Content

- Hard Disk (HDD)
- Disk Structure
- Disk Access
- Disk Scheduling
- RAID Structure

Hard Disk (HDD)

- A **hard disk drive** is a secondary storage device used to store data permanently in a computer.
  - Platter
  - Tracks
  - Sectors
  - Read/Write Head
  - Actuator Arm

Disk Structure



Disk Access – Key Components

- **Seek Time**
  The time taken for the disk head to move to the required track (cylinder).
- **Rotational Latency**
  The time taken for the disk to rotate so that the required sector comes under the read/write head.
- **Transfer Time**
  The time taken to read or write the data once the head is in the correct position.

Total Disk Access Time

$$T_{access} = T_{seek\ time} + T_{rotational\ latency} + T_{transfer\ time}$$

Disk Scheduling

- Disk scheduling is the method used by the operating system to decide the order in which disk read/write requests are serviced.
- The main goal is to **reduce disk access time** and improve overall system performance.

Why Disk Scheduling

- In an operating system, multiple processes often request access to the disk at the same time. These requests are placed in a **queue**, waiting to be processed.
- If the system services these requests inefficiently, the disk head will move unnecessarily, increasing access time and slowing down overall system performance. Therefore, disk scheduling algorithms are used to **optimize the order of requests** and improve speed.

Disk Scheduling (Cont.)

- Note that drive controllers have small buffers and can manage a queue of I/O requests (of varying "depth")
- Several algorithms exist to schedule the servicing of disk I/O requests
- The analysis is true for one or many platters
- We illustrate scheduling algorithms with a request queue (0–199)

```
98, 183, 37, 122, 14, 124, 65, 67
```

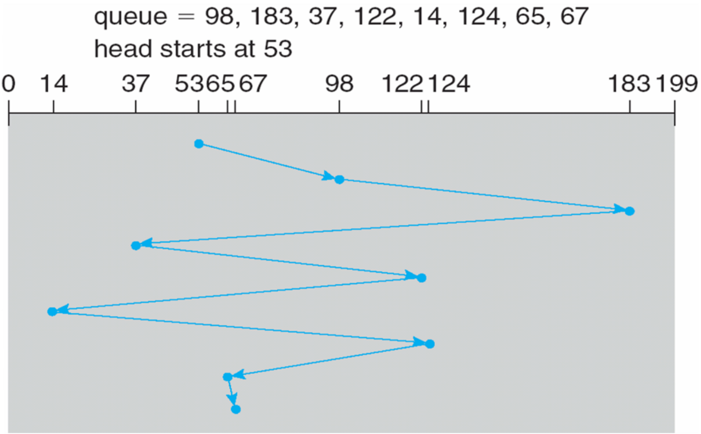
```
Head pointer 53
```

Disk Scheduling Algorithms

- FCFS Algorithm
- SSTF Algorithm
- SCAN Algorithm
- C-SCAN Algorithm
- LOOK Algorithm
- C-LOOK Algorithm

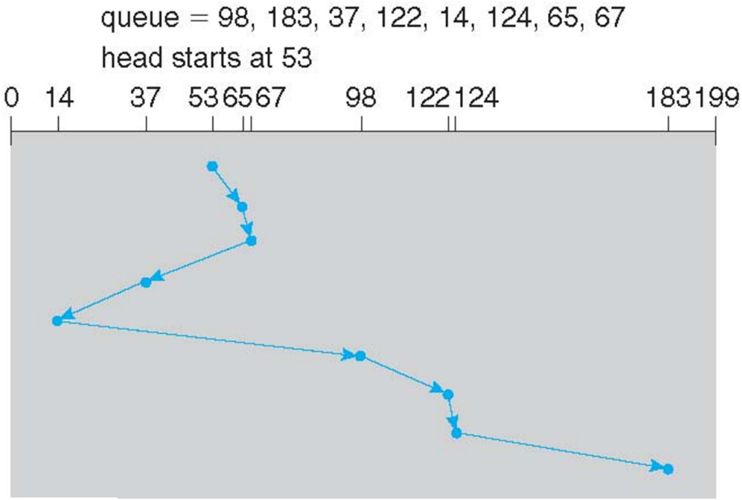
FCFS Algorithm

- The scheduling algorithm with a request queue (0–199)



SSTF Algorithm

- The scheduling algorithm with a request queue (0–199)

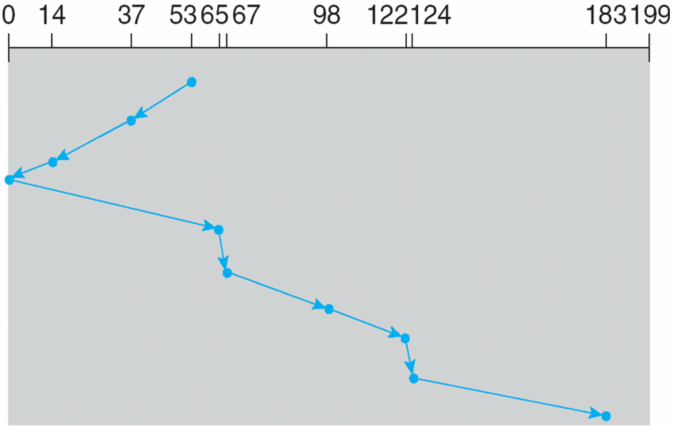


SCAN / elevator Algorithm

- The scheduling algorithm with a request queue (0–199)

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53, and initially moving to the left direction

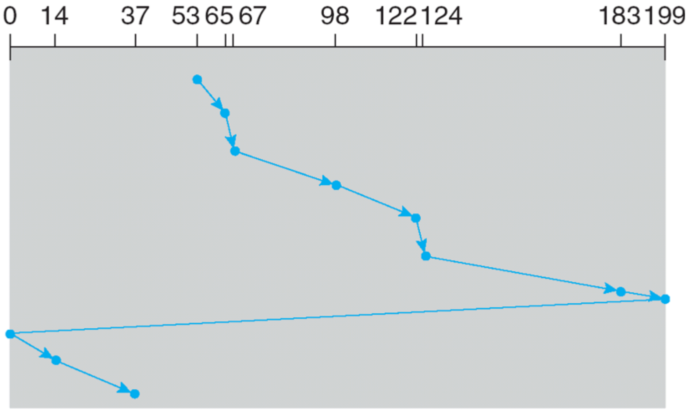


C-SCAN Algorithm

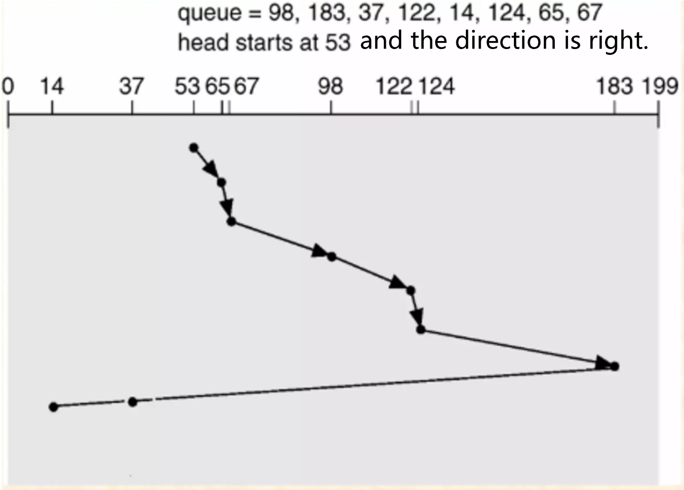
- The scheduling algorithm with a request queue (0–199)

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53 and previously it was at 36.



- The scheduling algorithm with a request queue (0–199)

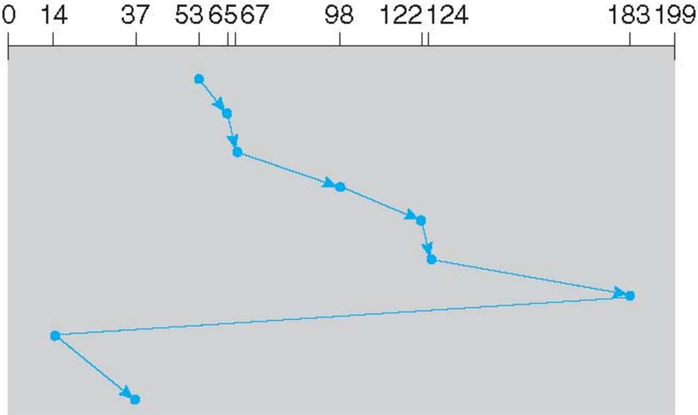


C-LOOK Algorithm

- The scheduling algorithm with a request queue (0–199)

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53 and the direction is right.



Redundant Array of Independent Disks (RAID)

- RAID is a storage technology that combines multiple physical hard disks into a single logical unit to improve:
  - Performance (speed)
  - Reliability (fault tolerance)
  - Storage capacity
- It allows data to be stored across multiple disks using techniques like striping, mirroring, and parity.

RAID 0 – Striping

- Data is divided into blocks and **distributed across multiple disks**
- Each disk stores part of the data, allowing **parallel access**

Advantages:

- Very high read/write performance
- Full use of total disk capacity

Disadvantages:

- No fault tolerance
- If one disk fails, **all data is lost**

(a) RAID 0: non-redundant striping.

RAID 1 – Mirroring

- Data is **duplicated on two or more disks**
- Each disk contains an exact copy of the data

Advantages:

- High reliability and data protection
- Easy data recovery if one disk fails

Disadvantages:

- Requires double storage capacity
- Higher cost

(b) RAID 1: mirrored disks.

RAID 5 – Striping with Parity

- Data and **parity information** are distributed across all disks
- Parity is used to **reconstruct data if one disk fails**

Advantages:

- Good balance between performance and reliability
- Efficient use of storage (less overhead than RAID 1)

Disadvantages:

- Write operations are slower due to parity calculation
- Can only tolerate **one disk failure**

(f) RAID 5: block-interleaved distributed parity.

RAID 6 – Double Parity

- Similar to RAID 5 but uses **two parity blocks**
- Can recover data even if **two disks fail**

Advantages:

- Very high fault tolerance
- Suitable for large storage systems

Disadvantages:

- Slower write performance than RAID 5
- More storage overhead

(g) RAID 6: P + Q redundancy.

RAID 10 (1+0) – Striping + Mirroring

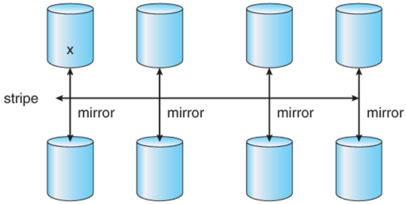
- Combines **RAID 1 (mirroring)** and **RAID 0 (striping)**
- Data is mirrored first, then striped across disks

Advantages:

- Very high performance
- High fault tolerance
- Fast recovery

Disadvantages:

- Requires many disks
- Expensive (50% storage overhead)



b) RAID 1 + 0 with a single disk failure.

End of Lecture 10

Thank You